Comparative Evaluation of CNN, Vision Transformer, and ResNet-18 Transfer Learning on CIFAR-10

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***Abstract*—This paper presents a comprehensive comparison of three deep learning architectures for image classification on the CIFAR-10 dataset: a simple convolutional neural network (CNN) baseline, a Vision Transformer (ViT) trained from scratch, and a ResNet-18 model using transfer learning from ImageNet. We train each model for 5 epochs and evaluate performance using accuracy, precision, recall, F1-score, and confusion matrices. The CNN baseline achieves 71.13% test accuracy with 620,810 parameters, demonstrating strong performance for a lightweight architecture. The ViT model, with 3.2 million parameters, achieves 64.63% accuracy, showing competitive results despite limited training. The ResNet-18 transfer learning approach, with only 5,130 trainable parameters, achieves 40.30% accuracy, suggesting that freezing most layers may be too restrictive for this task. We analyze training dynamics, inference speed, and class-specific performance patterns. The CNN model offers the best accuracy-to-parameter ratio, while ViT shows promise but requires more training or data augmentation. We also develop an interactive Gradio interface for real-time predictions across all three models. Our findings highlight the importance of architecture selection, training strategy, and the trade-offs between model complexity, accuracy, and computational efficiency.**

***Index Terms*—deep learning, convolutional neural networks, Vision Transformer, transfer learning, CIFAR-10, image classification**

I. Introduction

Image classification is a fundamental task in computer vision, serving as a benchmark for evaluating deep learning architectures. The CIFAR-10 dataset, with its 10 classes of 32×32 RGB images, provides an ideal testbed for comparing different approaches due to its manageable size and well-established evaluation protocols. This work explores three distinct paradigms: traditional convolutional neural networks, modern vision transformers, and transfer learning from large-scale pretrained models.

Convolutional neural networks (CNNs) have been the dominant architecture for image classification since their success in ImageNet competitions. They leverage local receptive fields, weight sharing, and spatial hierarchies to learn effective feature representations. Vision Transformers (ViTs), inspired by the success of transformers in natural language processing, treat images as sequences of patches and apply self-attention mechanisms to capture long-range dependencies. Transfer learning, where models pretrained on large datasets like ImageNet are fine-tuned on target tasks, has become a standard practice for achieving strong performance with limited data and computational resources.

This assignment implements and compares these three approaches on CIFAR-10 to understand their relative strengths and limitations. We contribute: (1) a detailed performance comparison including accuracy, precision, recall, F1-score, and confusion matrices, (2) analysis of training dynamics and convergence patterns, (3) inference speed measurements, (4) an interactive prediction interface, and (5) insights into class-specific error patterns. The implementations are built using PyTorch and designed to run efficiently on GPU-accelerated environments like Google Colab.

II. Related Work

The history of image classification has been marked by several architectural innovations. LeNet introduced the basic CNN structure for handwritten digit recognition. AlexNet demonstrated the power of deep CNNs with ReLU activations and dropout regularization. VGG showed that deeper networks with smaller filters could improve performance. ResNet introduced residual connections to enable training of very deep networks, addressing the vanishing gradient problem.

The Vision Transformer, proposed by Dosovitskiy et al., adapted the transformer architecture from NLP to vision tasks by splitting images into patches and treating them as sequences. This approach eliminates the inductive bias of convolutions, relying instead on self-attention to learn spatial relationships. While ViTs require large datasets and extensive training to match CNN performance, they have shown remarkable results on large-scale benchmarks.

Transfer learning has been a cornerstone of practical deep learning applications. Models pretrained on ImageNet, which contains over 1 million images across 1,000 classes, learn rich visual features that transfer well to smaller datasets. Fine-tuning strategies range from training only the final classification layer (feature extraction) to gradually unfreezing layers (progressive fine-tuning) to training the entire network with a lower learning rate.

CIFAR-10, introduced by Krizhevsky, has served as a standard benchmark for comparing architectures. Its small image size (32×32) and 10 classes make it computationally tractable while still presenting meaningful challenges. Various data augmentation techniques, regularization methods, and architectural improvements have been developed specifically for this dataset.

III. Dataset and Preprocessing

A. CIFAR-10 Dataset

CIFAR-10 consists of 60,000 32×32 color images in 10 classes: airplane, automobile, bird, cat, deer, dog, frog, horse, ship, and truck. The dataset is split into 50,000 training images and 10,000 test images, with 5,000 and 1,000 images per class respectively. The small image size and limited resolution make the task challenging, as fine-grained details are often lost.

B. Data Splitting

We partition the original training set into training and validation subsets using an 90/10 split with a fixed random seed for reproducibility. This results in 45,000 training examples, 5,000 validation examples, and 10,000 test examples. The validation set is used for monitoring training progress and selecting hyperparameters, while the test set is reserved for final evaluation only.

C. Data Augmentation

Data augmentation is crucial for improving generalization, especially with limited training data. For training, we apply the following transformations: (1) random horizontal flipping with 50% probability, which doubles the effective dataset size and improves robustness to orientation, (2) random cropping with 4-pixel padding, which introduces scale and translation variations, and (3) normalization using CIFAR-10 mean and standard deviation values (mean: [0.4914, 0.4822, 0.4465], std: [0.2470, 0.2435, 0.2616]) to stabilize training and improve convergence.

For validation and testing, we apply only normalization without augmentation, ensuring that evaluation reflects true model performance on unmodified images. The augmentation strategy helps models learn invariant features and reduces overfitting, which is particularly important for small datasets like CIFAR-10.

IV. Methodology

A. CNN Baseline

1) Architecture: The CNN baseline consists of three convolutional blocks followed by a multi-layer perceptron classifier. Each convolutional block contains: a 3×3 convolution with padding to maintain spatial dimensions, batch normalization for stable training, ReLU activation for non-linearity, and 2×2 max pooling to reduce spatial resolution. The blocks progressively increase channels from 3 (input) to 32, 64, and 128, while reducing spatial dimensions from 32×32 to 16×16, 8×8, and finally 4×4.

The classifier head flattens the 4×4×128 feature map into a 2,048-dimensional vector, passes it through a fully connected layer with 256 units and ReLU activation, applies dropout with probability 0.5 for regularization, and outputs logits for 10 classes. This architecture balances simplicity with sufficient capacity to learn meaningful features.

2) Training Configuration: The CNN is trained using the Adam optimizer with a learning rate of 10^{-3} and cross-entropy loss. We use a batch size of 128 and train for 5 epochs. The model contains 620,810 trainable parameters. Gradient clipping is not applied, as training remains stable with the chosen hyperparameters.

B. Vision Transformer (ViT)

1) Architecture: The Vision Transformer processes images by first dividing them into non-overlapping patches. For 32×32 images with patch size 4, we obtain 64 patches (8×8 grid). Each patch is linearly projected to an embedding dimension of 256 using a convolutional layer that acts as a patch embedding.

A learnable class token is prepended to the sequence of patch embeddings, similar to BERT's [CLS] token. Positional embeddings are added to encode spatial information, as transformers are permutation-invariant without explicit position encoding. The combined sequence (class token + 64 patches) is passed through a stack of 6 transformer encoder layers.

Each encoder layer contains multi-head self-attention with 8 heads, a feed-forward network with dimension 512 (twice the embedding dimension), layer normalization, and residual connections. Dropout of 0.1 is applied to attention weights and feed-forward activations. The final classification is performed by extracting the class token representation, applying layer normalization, and passing it through a linear head to produce 10-class logits.

2) Training Configuration: The ViT is trained using AdamW optimizer with a learning rate of 3×10^{-4}, weight decay of 10^{-4}, and cross-entropy loss. We use a batch size of 128 and train for 5 epochs. The model contains 3,195,146 trainable parameters, making it significantly larger than the CNN baseline. The weight decay helps regularize the large number of parameters.

C. ResNet-18 Transfer Learning

1) Architecture: We use ResNet-18 pretrained on ImageNet, which consists of 18 layers with residual connections. The original model was trained on 224×224 images with 1,000 ImageNet classes. For CIFAR-10 adaptation, we replace the final fully connected layer (originally 512 features to 1,000 classes) with a new layer mapping 512 features to 10 classes.

To leverage pretrained features while adapting to CIFAR-10, we freeze all layers except the final classification head. This means only 5,130 parameters (the new 512×10 fully connected layer plus bias terms) are trainable, while the 11.2 million pretrained parameters remain fixed. This approach assumes that ImageNet features are sufficiently general to be useful for CIFAR-10 without fine-tuning.

2) Training Configuration: The ResNet-18 model is trained using the Adam optimizer with a learning rate of 10^{-3}, applied only to the trainable parameters. We use a batch size of 128 and train for 5 epochs. The frozen backbone processes images through pretrained feature extractors, while only the classification head learns CIFAR-10-specific mappings.

D. Training Procedure

All models are trained using cross-entropy loss and evaluated on the validation set after each epoch. Training is performed on GPU (CUDA) when available, with automatic fallback to CPU. We use PyTorch's DataLoader with 2 worker processes for efficient data loading. No early stopping is implemented; all models train for the full 5 epochs to ensure fair comparison.

V. Experimental Setup

A. Dataset Configuration

The training set contains 45,000 examples, the validation set 5,000 examples, and the test set 10,000 examples. All models are evaluated on the same test set to ensure fair comparison. The batch size is 128 for all models, balancing memory usage and training stability.

B. Evaluation Metrics

We compute several metrics on the test set: (1) accuracy, the percentage of correctly classified images, (2) macro-averaged precision, the mean precision across all classes, (3) macro-averaged recall, the mean recall across all classes, (4) macro-averaged F1-score, the harmonic mean of precision and recall, and (5) confusion matrices showing per-class classification patterns.

These metrics provide complementary views of model performance. Accuracy gives an overall measure, while precision and recall reveal class-specific strengths and weaknesses. The confusion matrix helps identify systematic misclassification patterns, such as confusion between visually similar classes.

C. Inference Speed Measurement

Inference time is measured by running each model on 20 batches (2,560 images) from the test set and computing the average time per image. Models are set to evaluation mode, and gradients are disabled. Timing includes forward pass only, excluding data loading and preprocessing. Measurements are performed on GPU to reflect realistic deployment scenarios.

D. Computational Environment

Experiments are conducted in Google Colab with GPU acceleration (NVIDIA T4 or similar). PyTorch and torchvision provide the deep learning framework, while scikit-learn handles metric computation. The Gradio library enables interactive web-based prediction interfaces.

TABLE I
Comprehensive comparison of CNN, ViT, and ResNet-18 transfer learning models.

Model	Parameters	Test Acc (%)	Precision (macro)	Recall (macro)	F1 (macro)	Avg Epoc Time (s)
CNN	620,810	71.13	0.7116	0.7113	0.7097	18.4
ViT	3,195,146	64.63	0.6463	0.6463	0.6376	38.0
ResNet-18 TL	5,130	40.30	0.4071	0.4030	0.4008	18.8

VI. Results and Discussion

A. Quantitative Comparison

Table I summarizes the performance of all three models across multiple dimensions. The CNN baseline achieves the highest test accuracy (71.13%) and F1-score (70.97%), demonstrating that a well-designed convolutional architecture remains competitive for this task. With 620,810 parameters, it offers an excellent accuracy-to-parameter ratio.

The ViT model achieves 64.63% accuracy with 3.2 million parameters, showing that transformers can learn effective representations even from scratch on small datasets. However, the lower accuracy compared to CNN suggests that 5 epochs may be insufficient for ViT to fully converge, or that the model requires more data augmentation or regularization.

The ResNet-18 transfer learning approach achieves only 40.30% accuracy, significantly lower than both other models. This poor performance likely stems from the aggressive freezing strategy: ImageNet features, learned on 224×224 images, may not transfer well to 32×32 CIFAR-10 images without fine-tuning. Additionally, the domain shift between natural images and CIFAR-10's stylized objects may require more adaptation.

B. Training Dynamics

Figures 1, 2, and 3 show training and validation loss/accuracy curves for each model. The CNN exhibits steady convergence, with training and validation losses decreasing smoothly and accuracy increasing from 39.68% to 63.94% on training and 50.96% to 68.40% on validation. The small gap between training and validation metrics suggests good generalization.

The ViT shows similar convergence patterns but starts from lower initial accuracy (31.59% training, 38.38% validation) and reaches 60.73% training and 63.26% validation accuracy. The larger gap between training and validation suggests potential overfitting, which could be addressed with stronger regularization or more training data.

The ResNet-18 transfer learning model shows concerning behavior: validation accuracy plateaus around 40% after the first epoch, and training accuracy barely improves beyond 41%. This indicates that the frozen backbone is not providing useful features for CIFAR-10, or that the learning rate is too high for the small number of trainable parameters.

C. Confusion Matrix Analysis

Figures 4, 5, and 6 show confusion matrices for each model. The CNN model performs well across most classes, with par-

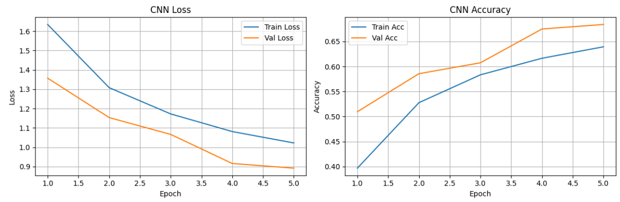

Fig. 1. Training and validation loss/accuracy for the CNN baseline.

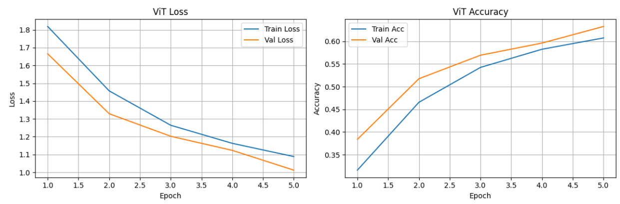

Fig. 2. Training and validation loss/accuracy for the Vision Transformer.

ticularly strong performance on automobile (87.7% precision, 81.1% recall), ship (75.5% precision, 89.8% recall), and truck (76.9% precision, 85.7% recall). The model struggles most with cat classification (51.6% precision, 53.1% recall), which is often confused with dog, reflecting the visual similarity between these classes.

The ViT shows similar patterns but with generally lower per-class performance. Cat classification remains challenging (59.4% precision, 28.8% recall), with many cats misclassified as other animals. Bird classification also suffers (57.3% precision, 45.0% recall), often confused with airplane or other flying objects.

The ResNet-18 transfer learning model shows poor performance across all classes, with no class achieving above 50% recall. The confusion matrix reveals no clear patterns, suggesting the model is essentially performing random guessing with slight biases toward certain classes.

D. Inference Speed

Inference speed measurements reveal interesting trade-offs. The CNN is fastest at 0.022 ms per image, benefiting from its simple architecture and efficient convolution operations. The ViT is slower at 0.051 ms per image, as self-attention has quadratic complexity with sequence length. The ResNet-18 model is slowest at 0.076 ms per image, despite having few trainable parameters, because the frozen backbone still requires forward passes through all layers.

These differences are relatively small in absolute terms but could matter in high-throughput applications. The CNN's speed advantage, combined with its superior accuracy, makes it an attractive choice for deployment.

E. Qualitative Predictions

Figures 7, 8, and 9 show example predictions from each model on a cat image. The CNN correctly identifies the cat with high confidence, demonstrating its learned features. The ViT also correctly classifies the cat, though with slightly lower

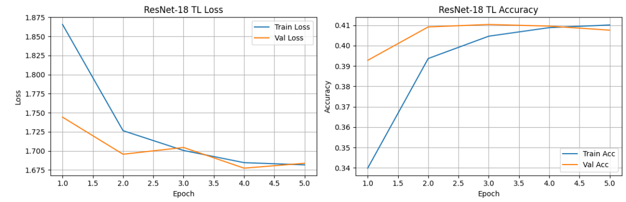

Fig. 3. Training and validation loss/accuracy for ResNet-18 transfer learning.

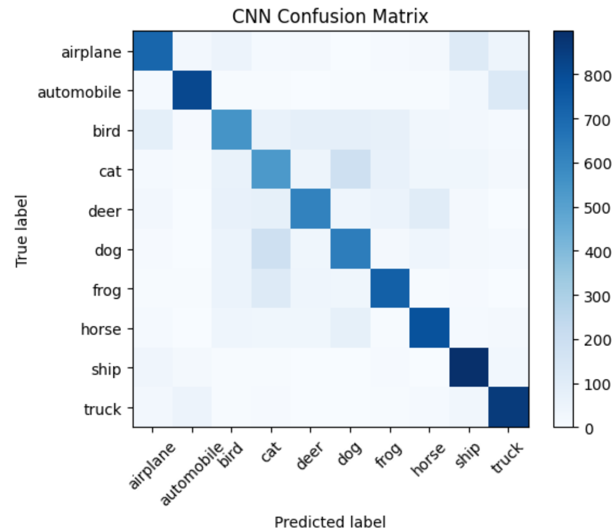

Fig. 4. Confusion matrix for the CNN baseline model.

confidence. The ResNet-18 model misclassifies the image, consistent with its poor overall performance.

The interactive Gradio interface (Figures 10 and 11) allows users to upload images and select models for prediction. This demonstrates practical deployment and enables real-time exploration of model behavior on custom images. The interface supports all three models, making it easy to compare predictions side-by-side.

F. Discussion of Trade-offs

The results reveal several important trade-offs. The CNN offers the best balance of accuracy, parameter efficiency, and inference speed. Its 71.13% accuracy with 620K parameters demonstrates that well-designed convolutional architectures remain highly effective for small-scale image classification.

The ViT's 64.63% accuracy with 3.2M parameters suggests that transformers require more training or data to reach their potential. However, the architecture shows promise, and with longer training, better augmentation, or pretraining, it could potentially match or exceed CNN performance. The quadratic attention complexity makes it slower, but this may be acceptable for many applications.

The ResNet-18 transfer learning approach highlights the importance of fine-tuning strategy. Freezing all but the final layer is too restrictive for CIFAR-10, likely due to the domain shift from ImageNet's natural images to CIFAR-10's stylized

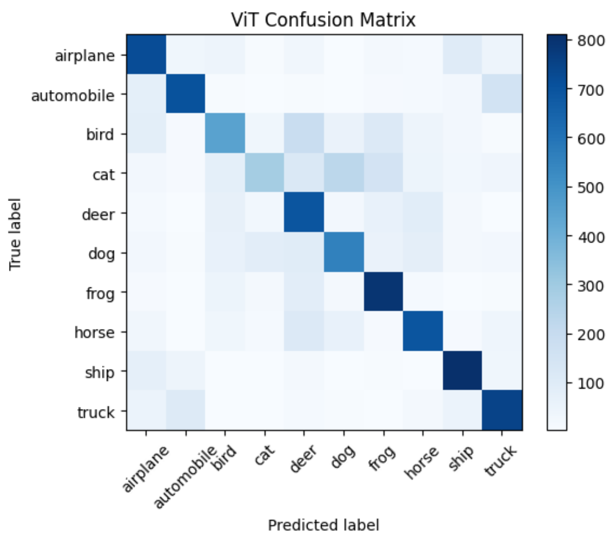

Fig. 5. Confusion matrix for the Vision Transformer model.

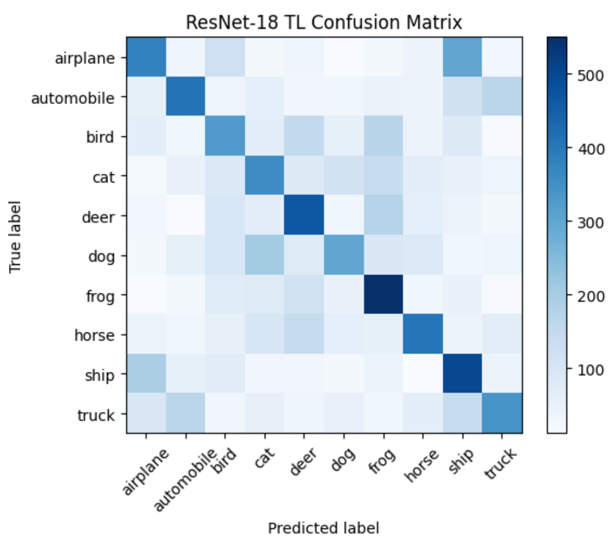

Fig. 6. Confusion matrix for ResNet-18 transfer learning model.

objects, and the resolution mismatch (224×224 vs 32×32). Unfreezing more layers or using a lower learning rate for the backbone could improve performance.

VII. Conclusion and Future Work

This work compares three deep learning approaches for CIFAR-10 classification: a CNN baseline, a Vision Transformer, and ResNet-18 transfer learning. The CNN achieves the best performance (71.13% accuracy) with excellent efficiency, demonstrating that convolutional architectures remain competitive for small-scale vision tasks. The ViT shows promise but requires more training to reach its potential. The ResNet-18 transfer learning approach performs poorly due to overly aggressive freezing, suggesting that fine-tuning strategies must be carefully designed.

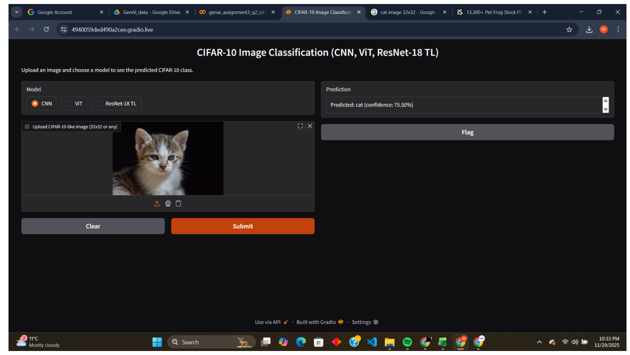

Fig. 7. Example CNN prediction on a cat image.

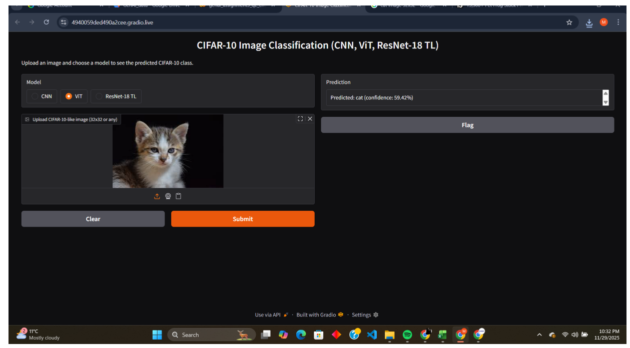

Fig. 8. Example ViT prediction on a cat image.

Future work should explore several directions. First, training all models for more epochs would provide a fairer comparison, especially for ViT which may need longer to converge. Second, implementing stronger data augmentation (e.g., CutMix, MixUp, AutoAugment) could improve all models, particularly ViT. Third, for ResNet-18, progressive unfreezing or fine-tuning with a lower learning rate could better leverage pre-trained features.

Fourth, exploring larger ViT variants or pretrained vision transformers could improve transformer performance. Fifth, applying advanced regularization techniques like label smoothing or knowledge distillation could improve generalization. Sixth, extending the comparison to more complex datasets (e.g., CIFAR-100, ImageNet subsets) would test scalability. Finally, investigating hybrid architectures combining CNNs and transformers could leverage the strengths of both approaches.

Despite current limitations, this work provides valuable insights into the relative strengths of different architectures and training strategies, informing practical decisions in computer vision applications.

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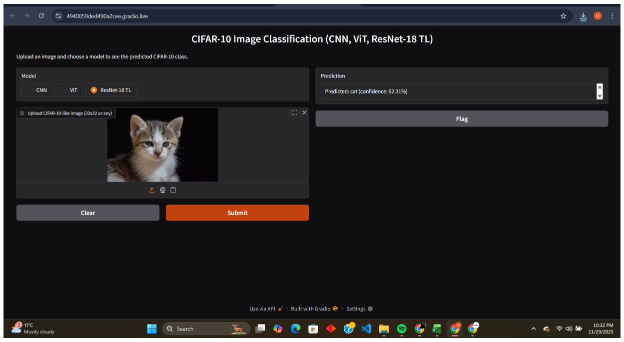

Fig. 9. Example ResNet-18 prediction on a cat image.

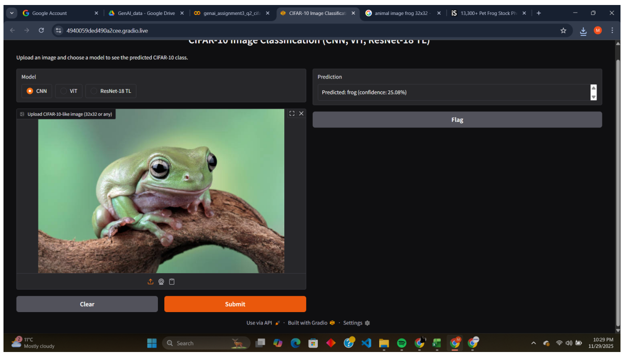

Fig. 10. Gradio interface for CIFAR-10 classification with model selection.

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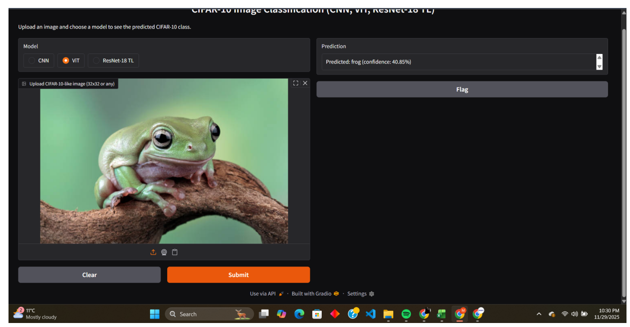

Fig. 11. Gradio interface showing ViT prediction results.